From Demonstrations to Deployment: Integrating AI, Control and Hardware in Wheeled Humanoids

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Abstract

Wheeled humanoids bring mobile manipulation to environments designed for people, though the full potential of a learned policy relies on a well-integrated system. Its performance depends on how task intent, demonstrations, perception, motion generation, sensing and control come together on a physical robot. This talk examines that integration challenge through the development and deployment of wheeled humanoid systems. Following a representative manipulation task, the talk traces the path from teleoperation and data collection through imitation learning and reinforcement learning to execution on hardware. It explores how robot and actuator integration, kinematics, motion planning, compliant control, simulation, calibration and fault handling help make learned behavior physically feasible, responsive and repeatable. Drawing on experience in robot controls and systems integration, and collaboration with teleoperation, data collection and AI teams, the talk will consider how the interaction between a policy and its hardware shapes development and evaluation. The talk concludes with practical considerations for mobile manipulation in real workflows: safe physical interaction, consistent behavior across robots, and systematic evaluation. It offers an engineering perspective on the work required to translate AI capability into useful robotic behavior.

Biosketch

Anubhav Dogra is a Senior Autonomy Engineer in Controls & AI at Humanoid in London, UK. Building on a background in mechanical engineering, a PhD in robotics from Indian Institute of Technology (IIT) Ropar, India and Post Doctorate Research Associate role at the University of Warwick UK, his work focuses on the critical integration of software and physical hardware. He bridges the gap between learned AI policies and control systems in wheeled and bipedal humanoids, specializing in robot bring-up, actuator and sensor integration. His engineering footprint spans for full stack robotics from low level to high level including whole-body kinematics and dynamics, simulation development, and the implementation of controllers that ensure safe, repeatable physical interaction in industrial deployments. Beyond the terminal and the robotics lab, Anubhav plays guitar and is an avid gamer drawn to expansive, narrative-driven action-adventures. He is also a budding coffee enthusiast, an active traveler, and is deeply interested in spiritual knowledge and personal expansion.